

Malhar Gudekar

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EDUCATION

University of Illinois Urbana Champaign

August 2024 - May 2026

Master of Science, Information Management

GPA: 3.8/4

• Coursework: Web Programming, Machine Learning & Cloud, Information Management, Information Consulting, Methods of Data Science

University of Mumbai

June 2019 - May 2023

Bachelor of Engineering, Computer Science

GPA: 9.25/10

• Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Object-Oriented Programming, Machine Learning, NLP

SKILLS

• **Languages:** Python, SQL, Java(leetcode), C++(leetcode)

• **Data Engineering:** ETL/ELT Pipelines, Data Modeling, Snowflake, PySpark, Workflow Orchestration (Airflow, Step Functions, AWS Glue)

• **Cloud & Tools:** AWS (S3, EC2, EKS), PostgreSQL, Neo4j, Git, CI/CD, Docker

WORK EXPERIENCE

University of Illinois Urbana Champaign

May 2025 - Present

Research Assistant - iSchool

Champaign, USA

- Built and maintained **AWS-based scalable data pipelines** to ingest high-volume wearable data for 100+ users, enabling distributed storage in **S3** and downstream analytics consumption
- Designed and optimized dimensional data models, star schemas and SCD Type 2 slowly changing dimensions in PostgreSQL and Snowflake, applying partitioning and indexing strategies to improve analytical query performance by 38% across high-volume production workloads
- Orchestrated data transformation workflows using Airflow DAGs and modular dbt models, implementing data quality checks, freshness assertions, null checks, referential integrity along with retry logic and alerting to ensure pipeline reliability and SLA compliance

Research Assistant - CHI AI in Infodemic Management

January 2025 - May 2025

- Designed and deployed scalable ELT pipelines in Python and dbt following a bronze/silver/gold medallion architecture, structuring raw, cleansed, and curated layers for high-volume social media datasets, reducing processing latency by 41% in distributed environments
- Built distributed real-time ingestion pipelines using **Kafka** and **Neo4j**, enabling streaming data processing, transformation, and graph-based analytics at scale
- Applied **distributed data processing and big data principles** to handle large-scale streaming data and ensure consistent data availability across systems

Business Intelligence Group

August 2025 - December 2025

Technical Consultant

Champaign, USA

- Built and deployed scalable **data processing pipelines** using **FastAPI**, **PostgreSQL**, and optimized storage layers to support large-scale analytics and downstream applications
- Designed fault-tolerant **ETL workflows** for ingesting and transforming high-volume document datasets, ensuring low-latency and reliable data
- Developed backend services and used **Git-based CI/CD workflows** to enable reliable deployment and real-time querying across distributed and high-availability environments

Swift Mobil Software Solutions Provider

July 2023 - October 2023

Data Analyst

Mumbai, IN

- Built **data pipelines** feeding **Power BI dashboards** for logistics KPIs, reducing reporting time by **40%** and improving data availability for business decision-making
- Processed large-scale datasets using **Python** and **PySpark distributed processing**, implementing transformations and aggregations for performance monitoring
- Developed SQL-based data quality checks covering null validation, referential integrity, and distribution drift to reconcile multi-source operational datasets, improving data accuracy and accelerating root-cause analysis by 26%

Pscope Technologies Pvt. Ltd.

January 2023 - June 2023

Data Analyst

Mumbai, IN

- Built and maintained **SQL-based data pipelines** supporting **Power BI dashboards**, designing data warehouse models, writing complex joins and aggregations, and ensuring delivery of analytics-ready datasets across business functions
- Automated **data transformation workflows** using VBA, implementing validation checks, error handling, and scheduling logic to streamline large-scale data processing and cut reporting time by **42%**

Stu/Dio at Illinois

August 2025 - Present

Project Manager

Champaign, USA

- Led cross-functional team of **6** engineers to deliver data-driven improvements, analytics insights, and system tracking
- Managed **Agile workflows** using JIRA, tracking delivery, QA issues, and sprint progress to ensure timely and reliable releases

PROJECTS

Scalable ML System for Customer Risk Prediction

March 2025

UIUC Hackathon

- Built an end-to-end analytics pipeline using Python, Pandas, and dbt-style modular transformations to process large-scale time-series transaction data, structuring ingestion, transformation, and feature layers to produce analytics-ready datasets with documented lineage and quality checks
- Designed and implemented **scalable ETL and feature engineering pipelines** for large datasets, capturing behavioral trends and temporal patterns to improve model performance
- Developed a **data-driven decision framework**, incorporating pipeline outputs and risk thresholds to improve prediction precision by 20% and enable scalable analytics workflows